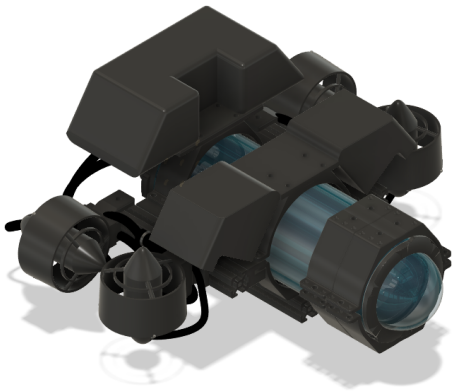
SubbyROV

Digital Twin to Real Vehicle Control Stack

User Guide



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1

Introduction

SubbyROV is a ROS 2 control stack for the QUT SubbyROV underwater vehicle. It was built so that the *same controller code* can drive both the Digital Twin (DT) and the real ROV, which speeds up autonomy development. It was made for the EGH490 Research Project at QUT.

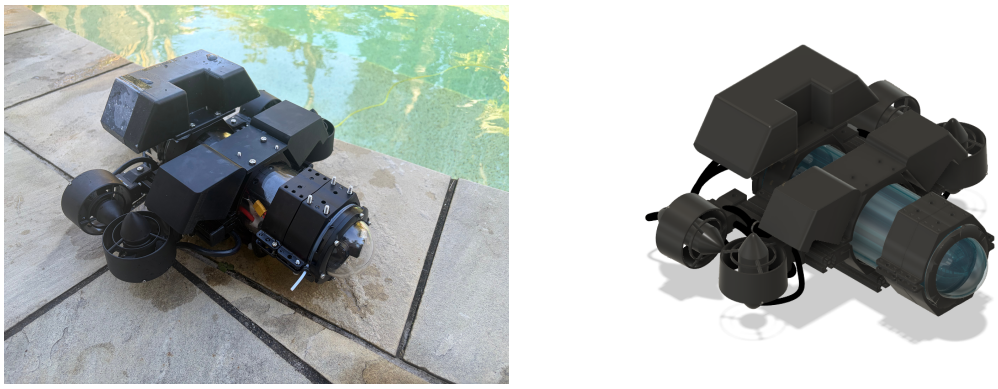

Figure 1.1. The real SubbyROV (left) and its Digital Twin (right).

1.1 About this guide

This guide covers what is not in the project README: the physical setup of the vehicle, how it is assembled and trimmed, and the upgrades recommended for whoever continues the project. Installation and run commands live in the README, and this guide links to the relevant section wherever they are needed.

> **See the README** The README is the source of truth for installation, run commands and troubleshooting: `https://github.com/Brad420169/QUT_ROV_Digital_Twin`

1.2 Digital Twin functions

The Digital Twin can perform these functions in a simulated 3D coral reef: teleoperation, station keeping, trajectory setting, and fish detection and tracking.

2

System Architecture

The shared controller publishes normalised surge, heave and yaw efforts on `/qut_rov/cmd_vel`. Whichever vehicle is selected, simulated or physical, turns those commands into thrust and reports depth and orientation back on `/qut_rov/depth` and `/qut_rov/imu`.

2.1 Shared control architecture

Figure 2.1 shows the shared control architecture used by both the Digital Twin and the physical vehicle.

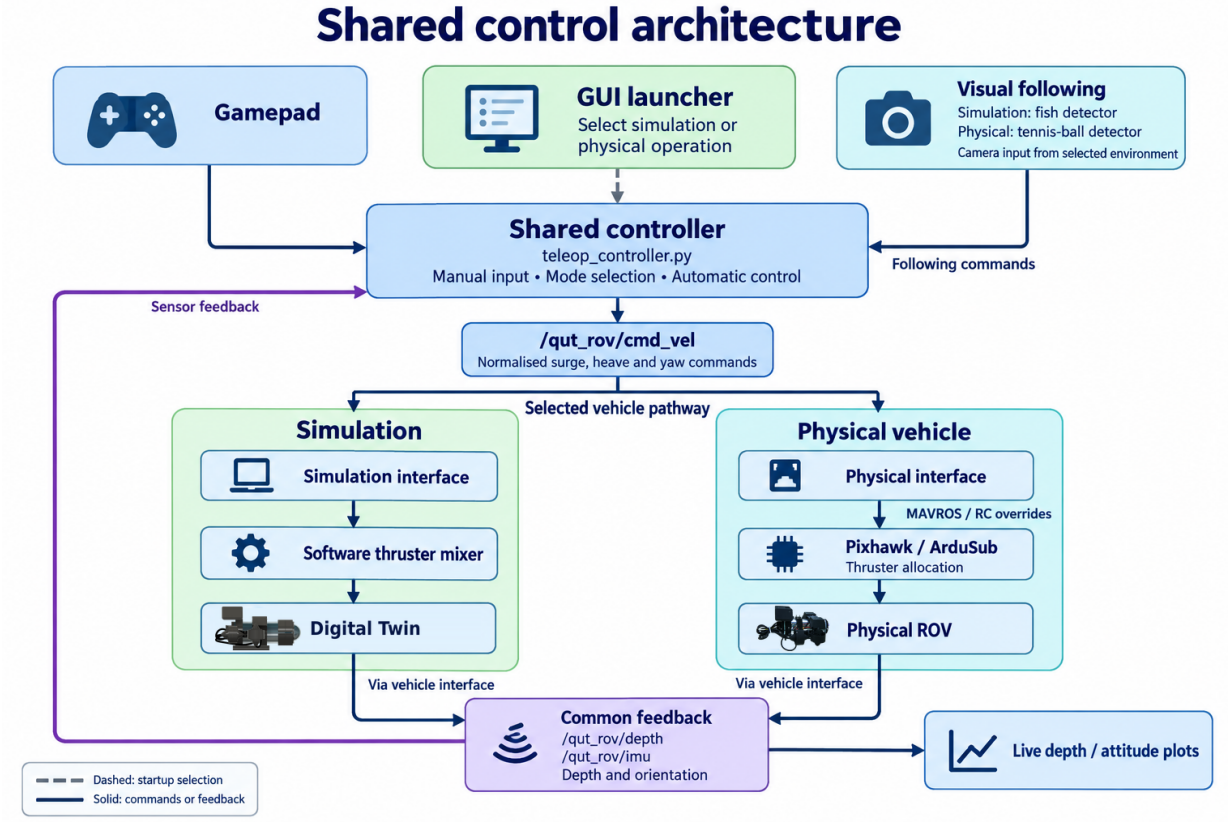

Figure 2.1. Shared control architecture.

2.2 Shared topics

Topic	Type	Description
`/qut_rov/cmd_vel`	`geometry_msgs/Twist`	Normalised surge, heave and yaw command
`/qut_rov/depth`	`std_msgs/Float64`	Vehicle depth
`/qut_rov/imu`	`sensor_msgs/Imu`	Orientation and rates

Table 2.1. Topics shared by the simulated and physical vehicle.

> **See the README** For what each node does and where the code lives, see the Nodes section of the README.

2.3 Thrust allocation

Thrust allocation is the one part of the pipeline that is *not* shared:

- **Digital Twin:** `thruster_mixer.py` converts surge, heave and yaw into four individual thruster commands in software.
- **Physical ROV:** commands go to ArduSub as RC channels, and the Pixhawk's SimpleROV-4 mixer does the allocation (channel 3 heave, channel 4 yaw, channel 5 surge).

Bringing these two in line is one of the recommended upgrades (see Match the simulator thruster mixer in Chapter 6).

3

Installation

Installation is fully covered by the README, and the scripts it describes do most of the work. It has three stages, all in one workspace checkout:

1. Install the real ROV software
2. Connect and set up the hardware
3. Install the Digital Twin simulation

> **Note** Before stage 2, connect the laptop to the ROV as described in Section 5.2. The hardware script needs the tether and Ethernet adapter in place.

4

Using the Digital Twin

Open **ROV** from the desktop or applications menu, select a vehicle model and choose **Launch simulation**.

> **See the README** Terminal and GUI-free launch commands are in the README under Verify the installation and start simulation.

4.1 Controls

Figure 4.1 shows the controller layout, which is the same for the Digital Twin and the real ROV.

Figure 4.1. SubbyROV controls.

5

Using the Real ROV

This chapter covers the physical side of SubbyROV: how it is powered, how the surface station is connected, how the enclosure is opened and sealed, and how the vehicle is trimmed.

5.1 Power architecture

Figure 5.1 shows the power and data architecture of the surface station and the submersible.

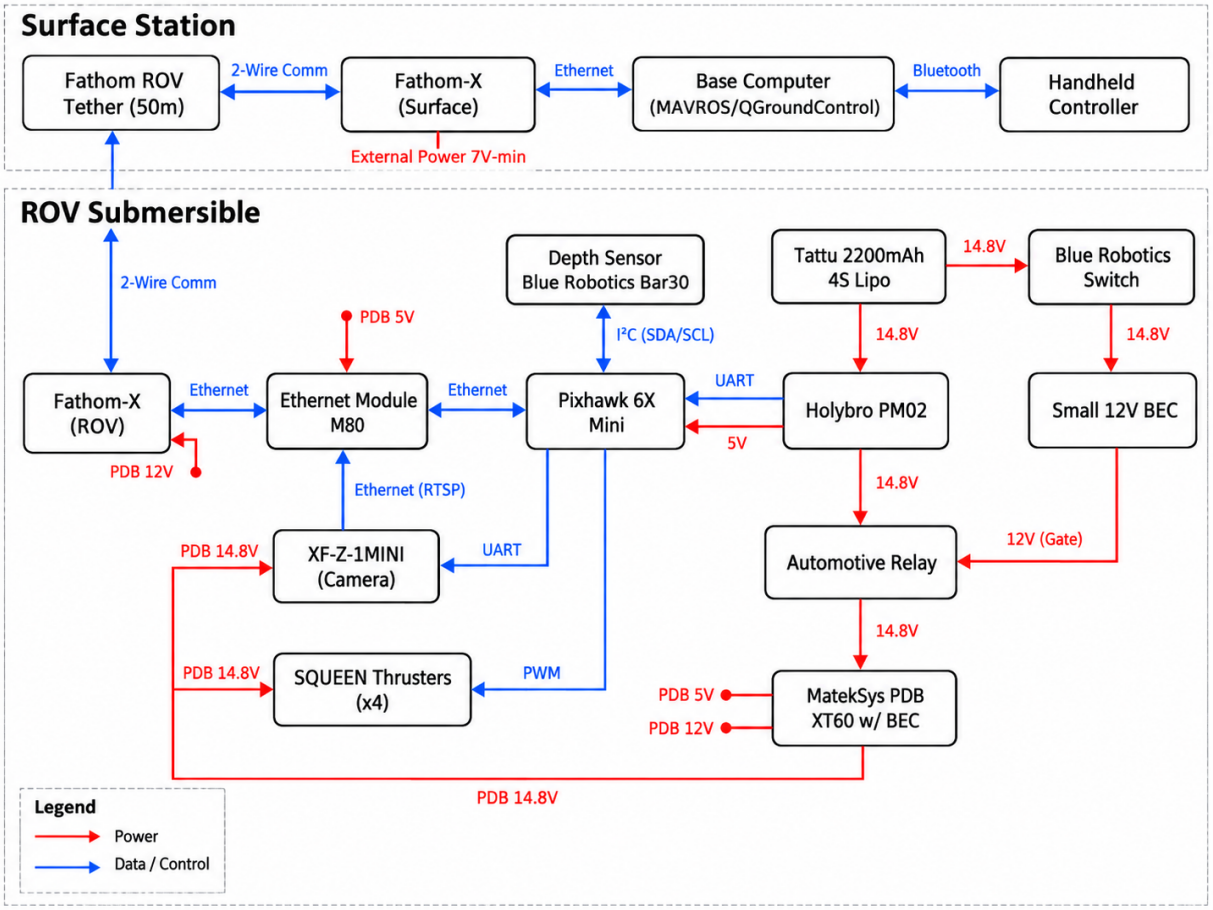

Figure 5.1. SubbyROV power and data architecture.

5.2 Operation setup

The laptop connects to a USB hub, which carries three connections:

- a **Bluetooth USB adapter** for the handheld controller;

- a **mini-USB cable** that powers the Fathom-X interface board;
- an **Ethernet adapter** that provides the data connection to the Fathom-X board.

Figure 5.2 shows the full setup.

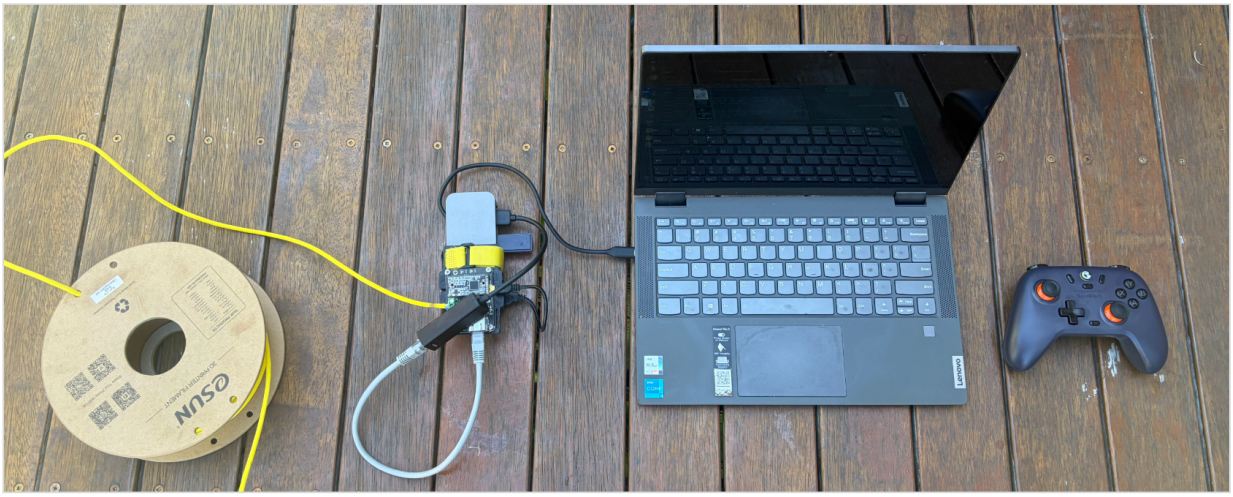

Figure 5.2. Operation setup: tether spool, Fathom-X interface board and USB hub, laptop and handheld controller.

5.2.1 Tether wiring

The tether connects to the Fathom-X board with **blue as positive** and **white as negative**. This matches the onboard configuration, so keep the same polarity at both ends.

5.2.2 Powering up

The ROV is powered up with the **kill switch** on the vehicle. The switch only powers the ROV if the battery, or another power input, is actually connected.

> ⚠ **Warning** If the ROV does not power up, check that the battery or other power input is connected before anything else.

5.3 Enclosure and end cap

The electronics are housed in a Blue Robotics watertight enclosure (WTE). The end cap seals it: the O-rings on the end cap flange press against the inside of the tube, sealing the ROV shut and keeping it watertight. A plastic lock holds the end cap in place.

5.3.1 Accessing the internals

1. Power the ROV down with the kill switch.
2. Move the plastic lock out of the way to release the end cap.
3. Pull the end cap straight out of the tube to reach the electronics.

5.3.2 Resealing

1. Check the O-rings and the inside of the tube are clean and undamaged. Any hair, sand or grit on an O-ring can cause a leak.

2. Push the end cap fully back into the tube.
3. Move the plastic lock back into place so the end cap cannot come loose.

> **Warning** Always check the end cap is fully seated and locked before putting the ROV in the water.

5.4 Buoyancy setup

The buoyancy blocks set the vehicle's trim. Set them up as shown in Figure 5.3:

1. Put the **bigger block** on the **rear mount**.
2. Put the **wings** on the **front mount**.
3. Shift the buoyancy configuration **towards the rear**.

Figure 5.3. Buoyancy block setup. The camera dome (front) is on the right; the bigger block sits on the rear mount and the wings on the front mount.

5.5 Running the ROV

Once the vehicle is connected, powered, sealed and trimmed, open **ROV** and select **Connect to real ROV**. Wait for the Pixhawk connection, depth calibration and camera to be ready before driving.

> **See the README** Full steps are in the README under Use the physical ROV.

6

Recommended Upgrades

For whoever continues this project

These are the changes that would most improve SubbyROV. The camera and visual tracking upgrades are linked: the current camera is what stopped visual tracking moving from the Digital Twin to the real vehicle. The last brings the Digital Twin's thrust allocation in line with the real ROV, so behaviour tuned in simulation transfers directly.

Reprint the buoyancy mounts in resin

Why The buoyancy mounts are currently printed in PLA. FDM prints have gaps between layers that can take on water, which slowly changes the vehicle's trim, and they are weaker along the layer lines.

What to do Reprint the buoyancy mounts in resin. Resin prints are solid and watertight, so the trim stays consistent between dives.

Replace the camera

Why The current camera overheats. At 80 °C it drops into a 5 fps thermal mode and the stack disables all vision control until it cools below 79.5 °C. This limits how long it can be used for anything that relies on the video feed.

What to do Replace it with the Blue Robotics Low-Light HD USB Camera, and design a mount for it that fits the existing WTE design.

Transfer visual tracking to the real ROV

Why Visual tracking works in the Digital Twin (fish detection and following) but could not be transferred to the physical vehicle. The current camera's overheating limit means vision control is cut out before tracking can run reliably.

What to do Once the camera is replaced (see Replace the camera), port the visual tracking to the real ROV using the new camera feed.

Match the simulator thruster mixer to the real ROV

Why The Digital Twin uses its own software mixer (`thruster_mixer.py`), while the real ROV uses the Pixhawk's ArduSub SimpleROV-4 mixer. Because they allocate thrust differently, the same command can move the two vehicles differently.

What to do Update the simulator's thruster mixer to match the physical thruster mixer, so the Digital Twin responds to commands the same way the real ROV does.

7

Troubleshooting

Software and install problems are covered in the README under Updates and troubleshooting. The items below are specific to the physical vehicle.

ROV does not power up
: Check that the battery or another power input is connected. The kill switch only powers the ROV when it has a power source (Section 5.2.2).

Vision control is disabled
: The camera has reached its temperature limit. At 80 °C it drops to 5 fps and all vision control is disabled until it cools below 79.5 °C. See Replace the camera in Chapter 6.

Water inside the enclosure
: Remove the ROV from the water and power it down immediately. Check the end cap O-rings for dirt or damage and make sure the end cap is fully seated and locked (Section 5.3).